QS-001 — Plant Site A Quality Management Standard

Rev 2 · Issued 10/01/2025 · automotive customer overlay (IATF 16949 aligned)

4.1 Context of the plant quality system

The plant shall determine the internal and external issues that affect its ability to deliver conforming product, including the condition of production equipment, the availability of experienced operators, and seasonal factors that alter process inputs. The quality system applies to all production and utilities equipment registered at Plant Site A and shall be reviewed when the equipment register changes. Interested parties include the customer whose overlay requirements this standard carries, the site maintenance organisation, and the regulatory authorities for the site. Where a customer-specific requirement conflicts with a base clause of this standard, the customer-specific requirement prevails and the conflict shall be recorded in the compliance mapping table maintained by the quality department.

7.1.5 Measurement traceability and calibration

All instruments used to control or verify product characteristics, including control valve positioners and their calibration equipment, shall carry current calibration status traceable to the site standard. An instrument found with expired calibration or an expired inspection record shall be treated as suspect: product controlled by it since the last valid record shall be evaluated for conformity, and the instrument shall not be used for acceptance decisions until calibration is restored. Stroke testing per SOP-108 constitutes the calibration record for control valves.

7.2 Competence and knowledge retention

The plant shall retain critical operational knowledge when experienced staff leave or retire. Undocumented practices that operators rely on shall be captured, reviewed against the written procedure, and either incorporated into the procedure or corrected. Where captured operator practice diverges from an approved procedure, the divergence shall be treated as a nonconformity of the documentation system and resolved through clause 10.2. Knowledge capture may use any recorded medium, including transcribed and translated spoken accounts from retiring operators; a captured claim is unverified experience until reviewed, and shall be stored separately from approved procedures so the two can never be confused at the point of use.

8.1 Operational planning and control

Production processes shall be planned and executed under controlled conditions, including approved procedures at the point of use, equipment maintained per the preventive schedule, and process parameters within their specified limits. Planned preventive maintenance intervals form part of controlled conditions; missed or deferred intervals on quality-critical equipment shall be recorded and assessed for product risk. Changes to operating conditions that could affect conformity, including seasonal changes to raw water quality and airborne debris load, shall be reviewed for their effect on planned intervals before the season begins, not after defects appear.

8.5.1 Control of production equipment condition

Production shall be carried out with equipment whose condition cannot degrade product conformity. For fluid transfer equipment, suction strainer condition and seal integrity are product-quality controls, not only maintenance concerns: a plugged strainer or passing seal can introduce contamination or pressure instability into the product stream. Cleaning and inspection intervals for such equipment shall reflect actual operating conditions, including seasonal debris load, and shall be reviewed when defect rates rise. Deviations from the published interval shall be recorded as nonconformities against this clause.

8.7 Control of nonconforming output

Product that does not conform to requirements shall be identified, segregated and dispositioned before release. Every nonconformity shall be recorded in the NCR register with the equipment or process implicated, the clause of this standard affected, and the disposition. Recurring nonconformities from the same equipment or cause shall escalate to root cause analysis under clause 10.2. Dispositions are limited to rework, reprocess, accept under concession with customer approval, or scrap; release of nonconforming product without a recorded disposition is a reportable quality system failure.

9.1 Process performance monitoring

Key quality indicators shall be computed monthly for each registered production process: process capability (Cpk) against specified limits and defect rate in parts per million (PPM). Indicators shall be computed from measured data held in the structured quality store; computed values, not narrative estimates, are the basis for decisions. A process with Cpk below 1.33 is not capable and shall not run unattended; a PPM value rising for three consecutive months triggers investigation regardless of absolute level. Indicator thresholds are defined per industry overlay in the standards pack configuration: parts per million defect rate for the automotive overlay carried by this site, with batch yield and lot traceability defined for other overlays. Computed indicators shall be traceable to the exact query and dataset used to produce them.

9.2 Internal audit

The quality system shall be audited internally at planned intervals. Audit findings shall reference the specific clause of this standard, the objective evidence observed, and the equipment or process concerned. Findings remain open until corrective action is verified effective by follow-up evidence, not by assertion.

10.2 Nonconformity and corrective action

For every nonconformity, the plant shall determine root cause before closing the record. Correction without root cause — repairing the same failure repeatedly, or re-aligning the same machine each quarter — does not satisfy this clause. Corrective action shall address the system that allowed the nonconformity: the procedure, the interval, the training, or the equipment condition, and its effectiveness shall be verified against subsequent performance indicators. Where operator practice has diverged from an approved procedure and the practice is found technically sound, the corrective action is a controlled revision of the procedure, not retraining of the operators to a written interval that field conditions contradict.